Examiner reports

Q1.

For many students part (a) was a straightforward application of a practised routine:

- $F = ma$ applied separately to each particle
- T eliminated from the two equations
- Use of $s = ut + \frac{1}{2}at^2$ to complete the solution.

A minority started with a single equation, $1.8g - 1.2g = ma$, which received zero marks. An equation separated from its underlying mechanical principle is hard to grasp, as evidenced by more of these students using $m = 1.8$ rather than $m = 3$.

Some students chose the wrong *suvat* equation and had to apply a second to find t. Premature or even final rounding sometimes lost the accuracy mark, where 3 significant figures were expected.

Students frequently approached the demand in part (b) in the wrong way. They should understand that what is required is an assumption that is necessary, sensible, specific and not already stated in the question. Thus saying that there was “no friction at the peg” or that “the string had negligible mass” scored zero marks. Similarly “no other forces act” is too general. A minority realised the necessity of sufficient length of string.

It is often better if the answer starts with an “I have assumed that” to ensure that the assumption goes on to be properly explained. For example “I have assumed that there is no air resistance” is better than just “air resistance” where the assumption is not clear.

Q2.

Nearly all students made good progress with part (a), with over half scoring full marks. Nearly all students successfully found the maximum possible friction and made a comparison with the applied horizontal force. The most frequently seen error was when students wrote a statement that was incorrect, often along the lines of “friction is greater than 150N so there is equilibrium”. This showed a misunderstanding of the friction model and could not be overlooked.

Part (b) was also well answered, but a lack of detail often meant that students had not fully justified their answer. Several impressive and detailed answers were seen, where students explained clearly that the horizontal component of the applied force was greater than the maximum friction so that the crate would not remain stationary.

Resolving of forces was generally successful.

Q3.

A high proportion of students had great difficulty in forming an appropriate equation of motion for part (a)(i). Those who attempted to treat the system as a whole made mistakes such as using the wrong total mass, including incorrect tension forces, using weight instead of mass or forgetting to include the resistance on the buggy. Some students considered the skater and buggy separately and similar errors were seen, however a fairly common error was to subtract the mass of the skater from the mass of the buggy in the equation of motion of the buggy.

Those students who wrote down a correct equation of motion usually completed the question successfully. Approximately half of the students scored full marks on part (a)(i).

In part (a)(ii) students were still able to make progress even if they had gone wrong in (a)(i). Provided their equation of motion was correct using their R they were given credit. Over half of students scored at least 2 marks.

Part (b) was a request for assumptions in a very standard model. A significant number of students gave assumptions that did not need to be made, as they were given in the question. For example, it was unnecessary to assume the road was flat as the question stated “horizontal road”. Students should also avoid words like “flat” or “straight” when they mean “horizontal”. A correct assumption would be that the rope was horizontal.

In part (c)(i), students who had gone wrong in earlier parts of the question were still able to make good progress and they were credited for using correct techniques. Correctly using their incorrect R from part (a)(ii) often resulted in 4/5 marks. Common errors were caused by students over complicating the problem, often attempting, unnecessarily, to find the time taken for either the buggy or skater to stop.

Over half of students made some progress on (c)(ii), but the general standard of explanation was poor. A significant number of students showed a misunderstanding of the forces model, or may have been trying to over-explain. Statements like “the rope has been dropped so the buggy no longer feels the resistance from the skater” are not accurate. It is more concise and accurate to simply say “there is no tension acting on the buggy”.

Q7.

While this question was generally done well there were more errors than in the first two questions. In part (a) the most common error was to include a force directed down the slope. Part (b) was done well, but a few students gave the answer 29.4 and a few used sin40° instead of cos40°. Part (c) was done very well and follow through marks were awarded to those with errors arising from part (b). Part (d) was a little more demanding, with the most common error being to work with only one force. There were many good answers to part (e), but some students gave answers that were not about the forces. For example, a statement like “The box is a particle.” did not relate to the forces acting on the box as was requested in the question.

Q8.

While there were many good solutions to this question, there were a number of errors in the solutions of some students. For example in part (a), some students ignored the tension in the tow bar acting on the tractor and created equations like $2500 - R = 3500 \times 0.2$. A number of students were able to find R correctly by finding the tension in the tow bar first before considering the forces on the tractor to find R. It was not uncommon to see students gain no marks in part (a), but gain full marks in part (b), when they considered the forces acting on the trailer correctly.
Many students realised that the answer for part (c) was expected to be the same as the answer to part (b) and simply stated this as expected. However, some worked with the forces acting on the tractor to find the tension and produced an equation rather than simply stating the value.

Q9.

Part (a) was a very standard question and many students did well with this part of the

question. However, a small number of students ignored the instruction to form two equations of motion. In part (b), some students did not convert the 40 cm to metres and obtained speeds that related to moving 40 m rather than 40 cm. These students did gain some marks, but lost the final accuracy mark.
In part (c) a common error was to use the wrong acceleration, with some students continuing to use 4.9 m s^{-2} rather than −9.8 m s^{-2}. In this part there was also some confusion between the values to use for u and v, with some students taking u as zero at the maximum height.
Most students were able to say that the acceleration would be reduced, when answering part (b), but a much smaller proportion of the students could give a good explanation. Those who mentioned the effect of the air resistance on the resultant force normally gained both marks.

Q10.

Part (a) was often done very well by students, with a reasonable number of students gaining all 11 marks. There were a few issues with the force diagram, with missing or additional forces. Many students seemed to have been helped by the answer that was given for the normal reaction and this in turn enabled many to find the friction correctly. Finding the acceleration was also done well, although there were a few sign errors and some students simply used 150 instead of 150 cos20°. Part (b) was much more demanding and there were far fewer correct responses. The most common error was not to take account of the changed angle when finding the normal reaction and to simply continue to use the value of 243 from part (a).
In part (c) there were some good responses, but a fair number of students incorrectly stated that the tension would be greater for a greater constant speed.

Q11.

Part (a) of this question was usually answered well. There were many correct solutions to part (b) but a few students tried to use the tensions found in part (a) forgetting that there was now an additional particle at B. A significant number of students did not include g in their calculations involving the particle of mass m kg and hence did not take the moment of a force.

Q12.

This question was generally done well. There were very many good force diagrams, with the most common error probably being the omission of arrow heads. Part (b) was done very well. While there were many good answers to part (b) some candidates simply calculated 4 × 3 = 12. There were also a few candidates who made a sign error, producing equations such as F – 50 = 12. Candidates were generally able to apply $F = \mu R$ to their answers from the previous parts, and many gained the follow through marks that were available. There were many good explanations for part (c), but also a number of confused answers, in which it was claimed that the coefficient of friction would have to increase.

Q13.

Candidates tended to do either very well or very badly on this question, although some were able to gain follow through marks in part (b) from an incorrect answer to part (a). Those candidates who realised that they had to consider the whole system in part (a) and did not include tensions usually obtained a correct solution. A large range of errors were seen which included adding a tension force, considering the weight or having the wrong multiple of R for the total resistance. The candidates seemed to find part (b) easier and were often able to form an equation, which they solved by just considering the car or the caravan. It was most common for them to consider the caravan, and most realised that the tension was given by T = 1600 + 2R.

Q14.

This question was done very well by candidates and there were many full correct solutions. The main source of errors in part (a) was due to using incorrect combinations of sin or cos and the angle that the candidates decided to work with. In part (b) some candidates used trigonometry while other used Pythagoras' Theorem. In the final part some candidates multiplied by 9.8 rather than dividing by it.

Q15.

The candidates who understood how to approach this question did very well, but others were unable to make much progress with this question.

Q16.

The candidates made very little progress with this question and did not seem to be able to deal with the resolving required.

Q17.

This question was done very well by candidates and there were many full correct solutions. The main source of errors in part (a) was due to using incorrect combinations of sin or cos and the angle that the candidates decided to work with. In part (b) some candidates used trigonometry while other used Pythagoras' Theorem. In the final part some candidates multiplied by 9.8 rather than dividing by it.

Q18.

Part (a) was done well by some candidates, but there were a significant number who proceeded as if this was a two particles connected by a string over a peg type of problem. Many times a pair of equations such as $18g - T = 18a$ and $T - 12g = 12a$ were seen. This made it impossible for candidates to complete this part of the question.
In part (b)(i), some candidates were able to get back on track, as they had the correct equation for the 18 kg particle, but some did use the acceleration from part (a) instead of 3 ms-2. In (b)(ii), there were many correct responses, but a few gave the answer 36 N, suggesting that there was some confusion between "reaction force" and "resultant force". The responses to part (b)(iii) were quite mixed with some good solutions. It was not uncommon for candidates to simply assume that the friction was equal to 36 N. Other candidates tried to find the difference between a "tension" which was often incorrect and the 36 N.
There were a number of good answers to part (c), but some were too vague. For example just writing "particles" was not enough as it did not make the connection that it was the block that was being modelled as a particle.

Q19.

The force diagram asked for in part (a) was generally well drawn, but some candidates did lose marks for not including arrow heads or appropriate labels. The candidates who

understood that they should resolve vertically to find the reaction force did well although some ended up with R = 78.4 + 0.5T or equivalent. Others produced very confused responses.
Those who had a correct expression for the normal reaction often made some progress in part (c). There were quite a number of candidates who made promising starts but then made algebraic errors when solving for T. There were a range of errors that were made in setting up an equation of motion. These included only using the friction force, using T instead of Tcos30° and omitting the ma term.

Q20.

Part (a), and in particular part (a)(i), caused problems for a number of candidates. The issue was that they did not consider the whole system or included a tension force in some inappropriate way. The candidates who found this difficult did not seem to appreciate the need to consider the whole system first. Some candidates did find the tension first and then considered the forces on the car to find P. Part (a)(ii) was done better than part (a)(i), but there was evidence of confusion as to which component of the system to consider and which forces were acting on it.

Part (b) was done very well and a high proportion of the candidates gained full marks on this part.

There were many good answers to part (c), but some candidates did make reference to the condition of the road surface and often included friction as if the car and caravan were sliding on the road surface.

Q21.

Part (a) was done successfully by the vast majority of the candidates. The force diagrams in part (b) were usually good, but the most common errors were to omit either the normal reaction or friction forces. Also some candidates used lines rather than arrows to represent the force.

In part (c), the candidates were often able to produce the printed answer, but in some cases did not show enough working to justify the award of full marks. For questions like this, candidates should be advised to show an answer to more than three significant figures before giving their final answer.

Those who started part (d) by resolving to find the friction force generally did well and completed this part of the question. A number of candidates did make significant errors by assuming, for example, that the friction force was $2g$ or in some cases $4g$.

Q22.

Parts (a), (b) and (c) were completed well by the majority of candidates and most of the errors were to be found in part (d). In part (a), some candidates made errors such as omitting arrowheads or using inappropriate labels, such as 4 kg instead of $4g$. In part (d), although there were many correct answers, the candidates produced a wide range of incorrect answers. Common errors included candidates' dividing either the 30 N (from the question) or 11.76 N (the friction) by the mass; other candidates calculated the resultant force and not the acceleration.

Q23.

Parts (a), (b) and (c) of this question were done very well. In part (a), almost all candidates used an equation of motion for each particle and there were very few 'single equation' solutions. In part (c), some candidates stated one incorrect assumption, for example one about the peg rather than the string. In part (d), a few candidates took the acceleration to be equal to g rather than the value obtained in the question.

Q24.

While there were a number of good solutions to this question, there were also a very significant number of weak attempts. In part (a), many candidates could write down either a correct horizontal or vertical equation, but then were unable to derive the required answer correctly. Some could get one equation but not the other. In some cases the equations were confused, for example containing both a and g in the same equation. In part (b), there were again a number of correct solutions. The most common error here was to work on the assumption that P would also be zero. The candidates who answered correctly produced a variety of solutions, which included finding an expression for Q and starting again as if there was only one string.

Q25.

Many candidates completed this question correctly. A common error occurring in part (a) was that some candidates in their diagram did not show that the two vertical upward reactions on the plank were different, or showed the two vertical gravitational forces acting at the same point. There were some interesting comments made in part (c), but most responses were correct.

Q26.

This question was also done well by the majority of candidates. While most candidates were able to produce the printed answer for part (b), there were some issues with the force diagram and finding the tension in part (c). The two main problems with the force diagram were including extra forces, for example friction or simply poor diagrams without arrows or labels. In part (c) there were some confused responses and some candidates tried to resolve the normal reaction from part (b).

Q27.

The candidates found this question more difficult than expected and there were many candidates who could not give correct responses to parts (c) and (d). While there were many good diagrams, some candidates did not label them clearly or did not include the 300 N force explicitly. The vast majority of candidates were able to gain all of the marks in part (b), but some did obtain the printed answer without sufficient working.

Part (c) was the most difficult part, with many candidates unsure which part of the train to consider. When candidates did consider the first carriage they often failed to include the 550 N force. The candidates did a little better on part (d), with those who considered the whole train generally being successful. Those who considered only the engine tended to omit the 1100 N force or not to use their answer from part (c).

Q28.

In part (a) many candidates lost marks for not including arrow heads or due to poor labelling of the forces. Several candidates did not show the two tensions as being equal on their diagrams. There were quite a few good attempts at part (b), with attempts at resolving vertically. Part (c) proved to be very challenging, although there were a number of good solutions. A number of candidates gained an M1 mark for the use of the friction law. The most common errors were to omit one force in their equation or to not include the 'ma' term in their equation of motion.

Q29.

Many candidates completed this question correctly. A common error occurred in part (a) where some candidates did not show, on their diagram, that the two vertical reactions on the plank were different. Creative algebra by a number of candidates was seen in part (b), particularly in the insertion of 'g' in the final answer.

Q30.

This question was found to be quite difficult by many of the candidates. In part (a) there were a number of incorrect force diagrams, including errors such as including a force acting directly down the slope. In parts (b) and (c) there were a lot of confused answers, mainly due to incorrect attempts at resolving. For example, in part (b), one error was to resolve the normal reaction into components instead of the weight. However, many candidates gained full marks for part (d), where follow-through marks were awarded when the candidates' values from parts (b) and (c) were used correctly in the friction law to find a coefficient of friction.

Q31.

The candidates generally did well with parts (a) and (b), which were fairly standard, but had more difficulties with the later parts. The main reasons that candidates lost marks in part (a) were because they made errors with the signs in the equations or because they did not form an equation for each particle. Those with sign errors in part (a) often did not obtain the correct tension in part (b).

In part (c), many errors were due to the candidates using the wrong values in the wrong places. For example, in part (c)(i), some used an acceleration of 9.8 instead of 1.96 while in part (c)(ii), some used an acceleration of 1.96 instead of 9.8. Similar confusion concerned the use of the distance of 4 metres in part (c)(i) instead of in part (c)(ii), while the time of 2 seconds was sometimes used in part (c)(ii) instead of in part (c)(i).

Q32.

This question was also done well by many candidates. There were relatively few errors on the force diagrams. In part (b)(ii), some candidates made errors, the most common being to multiply the 30 by 0.5 or subtract the 30 from 49. Part (c) caused difficulties for some candidates. The most common errors here were to attempt to include the weight or to omit the friction force.

Q33.

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In part (c), many errors were due to the candidates using the wrong values in the wrong places.

For example, in part (c)(i), some used an acceleration of 9.8 instead of 1.96, while in parts (c)(ii) and (c)(iii), some used an acceleration of 1.96 instead of 9.8. Similar confusion concerned the use of the distance of 4 metres in part (c)(i) instead of the later parts, while the time of 2 seconds was sometimes used in the later parts instead of in part (c)(i).

In part (c)(iii), the most common approaches involved the use of constant acceleration equations. While some candidates used the correct numerical values in their equations, many made errors with the signs. Another common approach was to find two times and add them together. With this method also, sign errors were often made by the candidates.

Q34.

There were many good solutions to part (a). Only a very small number of candidates ignored the instruction to form an equation of motion for each particle. A few candidates assumed that the heavier particle would accelerate upwards when released. Some of these candidates worked through to produce a negative acceleration and stated that magnitude would be the required value. However, some of the candidates who took this approach "lost" a negative sign in their working and ended up with a positive acceleration.

In part (b) there was one common error, which was seen on a fairly large number of scripts. This was to obtain the distance moved by each particle, but not to double this to give the distance between the particles. A very small number of candidates used an acceleration of 9.8 instead of 0.98.

Q35.

The printed answer in part (a) seemed to help the candidates as there were very few incorrect answers to this part of the question. In part (b), while there were many good responses, some candidates used Speed = $\frac{\text{Distance}}{\text{Time}} = \frac{0.9}{0.6} = 1.5$ and then applied the equation $v = u + at$ to obtain an acceleration of 2.5. There were some very good answers to part (c), but some candidates gave answers such as "The accelerations are different." but did not explain why. In the better explanations, candidates made clear references to friction forces or air resistance.